Exam AWS Certified Developer - Associate DVA-C02 All Questions

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EXAM AWS CERTIFIED DEVELOPER - ASSOCIATE DVA-C02 TOPIC 1 QUESTIO...

Exam question from Amazon's AWS Certified Developer - Associate DVA-C02
Question #: 129
Topic #: 1
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A developer is creating an AWS Lambda function. The Lambda function will consume messages from an Amazon Simple Queue Service (Amazon SQS) queue. The developer wants to integrate unit testing as part of the function's continuous integration and continuous delivery (CI/CD) process.

How can the developer unit test the function?

A. Create an AWS CloudFormation template that creates an SQS queue and deploys the Lambda function. Create a stack from the template during the CI/CD process. Invoke the deployed function. Verify the output.

B. Create an SQS event for tests. Use a test that consumes messages from the SQS queue during the function's CI/CD process.

C. Create an SQS queue for tests. Use this SQS queue in the application's unit test. Run the unit tests during the CI/CD process.

D. Use the aws lambda invoke command with a test event during the CIICD process.

Show Suggested Answer

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by patrick889 at *June 17, 2023, 6:32 a.m.*

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Comments

gagol14 Highly Voted 2 years, 9 months ago

Selected Answer: C

Unit testing is a type of testing that verifies the correctness of individual units of source code, typically functions or methods. When unit testing a Lambda function that interacts with Amazon SQS, you can create a separate test SQS queue that the Lambda function interacts with during testing. You would then validate the behavior of the function based on its interactions with the test queue. This approach isolates the function's behavior from the rest of the system, which is a key principle of unit testing.

Option A is incorrect because AWS CloudFormation is typically used for infrastructure deployment, not for unit testing.

Option B is incorrect because it does not actually test the function; it only creates an event.

Option D is incorrect because the 'aws lambda invoke' command is used to manually trigger a Lambda function, but doesn't necessarily facilitate testing the function's behavior when consuming messages from an SQS queue.

upvoted 16 times

redfivedog Highly Voted 2 years, 8 months ago

Selected Answer: D

D is correct here. Both B and C are integration tests as they are using an actual SQS queue in the tests and not mocking it out.

upvoted 13 times

mooncake1 Most Recent 1 year, 2 months ago

Selected Answer: D

Those who select C and still think they are right - Should study more about unit test & integrate testing before AWS

upvoted 1 times

sumanshu 1 year, 3 months ago

Selected Answer: D

A) Eliminated - This approach involves creating and destroying AWS resources (SQS, Lambda) for each test, which increases the time, complexity, and potential costs associated with running tests.

B) Eliminated - t involves real AWS resources (SQS)

C) Eliminated - Even though it's meant for testing, you're still using an actual AWS service (SQS) to conduct the test, making it less of a unit test and more of an integration test.

D) Correct - You use the aws lambda invoke command to invoke the

Lambda function directly during the test process, without needing to rely on actual resources like an SQS queue.

upvoted 1 times

ShakthiGCP 1 year, 3 months ago

Selected Answer: C

Unit testing is a type of testing that verifies the correctness of individual units of source code, typically functions or methods

upvoted 1 times

f271c23 1 year, 3 months ago

Selected Answer: C

option C seems more appropriate. As there is a clear isolation of test stage environment and it is staged prior to the deployment.
Option D triggered manually and consuming from the same queue that was meant for the Production environment

upvoted 1 times

CloudChingon 1 year, 4 months ago

Selected Answer: D

It's testing so option D seems more logic. The other options would put a message in SQS.

upvoted 1 times

Anandesh 1 year, 8 months ago

Selected Answer: D

https://docs.aws.amazon.com/lambda/latest/dg/testing-guide.html

upvoted 1 times

65703c1 1 year, 10 months ago

Selected Answer: D

D is the correct answer.

upvoted 1 times

Moralles 1 year, 11 months ago

Selected Answer: D

In the case of unit tests, whose objective is to isolate the tested unit, option D is the one that most isolates the unit.

upvoted 1 times

41eb566 2 years ago

Selected Answer: C

To unit test the AWS Lambda function that consumes messages from an Amazon SQS queue as part of the CI/CD process, the developer can follow option C:

C. Create an SQS queue for tests. Use this SQS queue in the application's unit test. Run the unit tests during the CI/CD process.

upvoted 1 times

KillThemWithKindness 2 years, 1 month ago

Selected Answer: D

In production, our Lambda function code will directly access the AWS resources we defined in our function handler; however, in our unit tests

we want to isolate our code and replace the AWS resources with simulations. This isolation facilitates running unit tests in an isolated environment to prevent accidental access to actual cloud resources.

https://aws.amazon.com/blogs/devops/unit-testing-aws-lambda-with-python-and-mock-aws-services/

upvoted 1 times

SerialiDr 2 years, 2 months ago

Selected Answer: D

D. Use the aws lambda invoke command with a test event during the CI/CD process: This option is closer to what unit testing entails. The aws lambda invoke command can be used to invoke the Lambda function with a simulated event payload that mimics an SQS message. This allows the developer to test the function's logic and handling of SQS messages without needing an actual SQS queue. The test can focus on how the function processes the input and generates output, which is the essence of unit testing.

upvoted 3 times

CrescentShared 2 years, 2 months ago

Anybody find this question in the exam, please? The question itself looks so wrong to me, the action of testing the lambda function does not seem like a 'unit test' already... Isn't the unit test testing all the Classes inside the lambda function?

upvoted 2 times

Certified101 2 years, 3 months ago

Selected Answer: C

C there should be a seperate isolated test enviroment
D will only invoke the lambda and not test SQS polling.

upvoted 2 times

tqiu654 2 years, 4 months ago

Selected Answer: D

ChatGPT:D

upvoted 1 times

ShawnWon 2 years, 4 months ago

B.
Option A (CloudFormation template for SQS queue and Lambda function) involves more of an integration test rather than a unit test. It's typically preferable to keep unit tests isolated and focused on the specific functionality of the function.

Option C (Create an SQS queue for tests) might involve additional setup and cleanup steps, and it could introduce dependencies that impact the isolation of unit tests.

Option D (aws lambda invoke command with a test event) is similar to Option B, but creating a test event is generally more flexible and allows for a clearer representation of the expected input to the Lambda function.

upvoted 3 times